DHANASEKARAN D
www.linkedin.com/in/dhanasekarand63

ALL INTERVIEW QUESTIONS TOPICWISE

AZURE DATA LAKE STORAGE:

1.What is Azure Storage and its Benefits?

2. How does Azure Blob Storage differ from Azure Data Lake Storage (ADLS Gen2)?

3. What are the different access tiers in Azure Blob Storage, and when should each be used?

4. Can you list and explain the different data storage services available in Azure?

5. How can data security be enforced in Azure Blob Storage?

6. What is Azure File Storage, and what are its main advantages?

7. Why is mounting storage recommended in Databricks instead of using access keys directly?

8. What are the best approaches to efficiently ingest large volumes of data into Azure Blob Storage?

9. What different blob types are available in Azure Blob Storage, and where are they used?

10. How do you securely connect Azure Databricks with Azure Data Lake Storage (ADLS)?

11. How can blob versioning be configured in Azure Blob Storage, and what advantages does it provide?

12. How would you improve the performance of a large-scale data lake that processes petabytes of data?

13. What parameters influence the pricing of an Azure Storage account?

14. How can you copy multiple file formats such as CSV, Parquet, and Excel using Azure Data Factory?

15. What is soft delete in Azure Blob Storage, and how is it enabled?

16. How can you design and apply a data lifecycle management strategy in Azure Data Lake Storage?

17. What are the different methods available to authorize access to an Azure Storage account?

18. How can you capture or log source file names while ingesting data using Azure Data Factory?

19. What type of data consistency does Azure Blob Storage provide, and how does it work?

20. How do you connect Azure Data Lake Storage with Azure Synapse Analytics to build an end-to-end analytics solution?

21. Can you explain what the AzCopy tool is and how it is used?

22. How do you improve data read performance from ADLS when using Apache Spark?

23. How can you monitor and diagnose performance-related issues in Azure Blob Storage?

24. How would you design a solution for near-real-time analytics on streaming data stored in Azure Data Lake Storage?

25. Why does an Azure Storage account provide two access keys instead of one?

26. What security mechanisms are available in Azure Data Lake Storage (ADLS)?

27. How can high availability be achieved in Azure Blob Storage using data replication options?

28. How would you design and orchestrate complex data pipelines that involve ADLS, Azure Data Factory, and multiple Azure services?

29. If you were designing a web application that allows users to upload images and large video files, which Azure storage service would you choose and why?

30. How would you configure redundancy in Azure Blob Storage to satisfy a defined availability SLA?

31. How would you plan and execute the migration of a large on-premises dataset to Azure Data Lake Storage?

32. As a data engineer, how would you provide temporary access to report files stored in Azure Blob Storage for external applications? However, you want this has to be accessible only for the next 7 days.

33. What is the role of the hierarchical namespace in Azure Data Lake Storage Gen2?

34. How do different blob types in Azure Blob Storage impact performance, and when should each be used?

35. How would you secure sensitive data stored in Azure Data Lake Storage (ADLS)?

36. If you are storing mission-critical data in an Azure Storage account, how would you ensure it is securely stored?

37. What factors would you consider when designing a GDPR-compliant data retention policy in Azure Blob Storage?

38. What strategies would you use to improve the performance of data processing in Azure Data Lake Storage?

39. Can you provide different access levels to users on an Azure Storage account, such as read-only for juniors and read/write for seniors? How would you implement it?

40. What are the key differences between Azure Data Lake Storage Gen1 and Gen2?

41. How would you implement blob versioning as part of a backup strategy, and how does it differ from soft delete?

42. How do you monitor and optimize data processing jobs in Azure Data Lake Storage and Databricks to ensure high performance?

43. How would you handle scenarios where multiple clients need to access the same blob simultaneously in Azure Blob Storage?

44. How would you handle large-scale data ingestion into Azure Data Lake Storage efficiently?

45. How would you implement encryption in Azure Blob Storage to comply with regulations across different regions or jurisdictions?

46. How would you handle real-time streaming data processing and store the results in Azure Data Lake Storage?

47. If your application is deployed in the US-East region, should the associated Azure Storage account also be in the same region? Why or why not?

48. How would you set up disaster recovery for an Azure Storage account to ensure data availability in case of regional failures?

49. How would you design a scalable and highly available solution for ingesting large volumes of unstructured data into Azure Blob Storage?

50. How would you design a solution to ingest and store large volumes of web server log files in Azure Data Lake Storage for analysis?

51. Can data in Azure Blob Storage be moved automatically between access tiers? Why would this functionality be useful?

52. How can you integrate Azure Data Lake Storage with Azure Synapse Analytics for querying and analyzing data?

53. How would you implement fine-grained access control for multiple applications accessing sensitive data in Azure Blob Storage?

54. How would you secure sensitive customer data stored in Azure Data Lake Storage to prevent unauthorized access?

55. Can you check how much your Azure Storage account is costing? If yes, how would you do it?

56. How would you implement data quality checks and validation for data ingested into Azure Data Lake Storage?

58. How would you archive infrequently accessed data in Azure Data Lake Storage to reduce storage costs?

59. How can you generate a detailed overview of containers, blobs, snapshots, and versions within an Azure Storage account?

60. How can you set ACLs in Azure Data Lake Storage to control access at the folder or file level?

61. How would you secure data in transit and at rest in Azure Blob Storage? To secure data in Azure Blob Storage:

62. For a project handling real-time streaming data, how would you integrate Azure Data Lake Storage into the solution?

63. If you want to share a checklist or cheat sheet publicly as a website or blog, how can you do this using an Azure Storage account?

64. How is data encrypted at rest and in transit in Azure Storage, and what options exist for sensitive workloads?

65. How can you host a static website using Azure Blob Storage, and what are the setup steps

66. How would you ensure GDPR compliance for sensitive data stored in Azure Data Lake Storage?

67. How would you deploy a simple HTML page as a static website using Azure Storage?

68. How can you integrate Azure Data Lake Storage with Databricks for efficient data processing?

69. How would you securely transfer data from on-premises systems to Azure Blob Storage?

70. How would you integrate Azure Data Lake Storage with on-premises data sources in a hybrid cloud setup?

71. You have two users: a junior team member and your manager. How would you assign read access to the junior and read/write access to the manager for a storage account?

72. What are some best practices to optimize costs for Azure Storage?

73. How do you configure CORS (Cross-Origin Resource Sharing) for Azure Blob Storage to allow a web application from another domain to access it

74. How would you set up monitoring and alerts for data operations in Azure Data Lake Storage (ADLS)?

75. If your storage account will hold mission-critical data, how would you extend the default soft delete retention period from 7 days to 90 days?

76. How do you handle schema evolution for data stored in Azure Data Lake Storage (ADLS)?

77. How would you plan and execute a large-scale data migration from another cloud provider to Azure Data Lake Storage (ADLS)?

78. Is it possible to move an existing Azure storage account to a different resource group? If yes, how would you do it?

79. How do you audit and log access to data stored in Azure Data Lake Storage (ADLS)?

80. With rapidly growing data in your ADLS, how would you design a partitioning strategy to optimize query performance and control costs?

81. If you initially selected Zone-Redundant Storage (ZRS) for a storage account instead of Geo-Redundant Storage (GRS), can you change the redundancy later? If yes, how?

82. If your pipeline reads data from ADLS, applies transformations, and writes back to ADLS, and you observe performance issues during transformations, how would you diagnose and resolve the bottleneck?

83. How would you enforce encryption for data stored in Azure Data Lake Storage, both at rest and in transit? Which Azure features and settings would you use?

84. How would you implement version control for files in Azure Data Lake Storage to enable rollback and auditability?

AZURE DATA FACTORY:

1. What is Azure Data Factory, and what are its primary use cases?

DHANASEKARAN D
www.linkedin.com/in/dhanasekarand63

2. How can you implement incremental data loading in Azure Data Factory to improve data movement efficiency?

3. How do you handle schema drift in Azure Data Factory when working with semi structured data sources?

4. How would you handle intermittent pipeline failures caused by network issues in Azure Data Factory?

5. What are pipelines, linked services, and datasets in Azure Data Factory?

6. What is the difference between full load and incremental load in Azure Data Factory?

7. How do you manage dependencies between multiple pipelines in Azure Data Factory to maintain the correct execution sequence?

8. How would you copy data from an on-premises SQL Server to Azure SQL Database on a daily basis using Azure Data Factory?

9. What are triggers in Azure Data Factory, and what types of triggers are available?

10. What are the different types of Integration Runtimes in Azure Data Factory?

11. How do you parameterize datasets and linked services in Azure Data Factory to make them reusable?

12. How would you transform data from a CSV file stored in Azure Blob Storage and load it into Azure SQL Database using Azure Data Factory?

13. What are Copy and Delete activities in Azure Data Factory, and how are they used?

14. How would you copy very large files (for example, 3 TB) from an on-premises system to Azure using Azure Data Factory?

15. How do you design error handling and retry mechanisms in Azure Data Factory pipelines to make them reliable?

16. How would you copy data from multiple CSV files in ADLS Gen2 to Azure SQL Database using Azure Data Factory?

17. What is the difference between Trigger execution and Debug execution in Azure Data Factory?

18. What is the Binary Copy method in Azure Data Factory, and when should it be used?

19. How do you integrate Azure Data Factory with Azure DevOps to enable CI/CD for data pipelines?

20. How do you ensure that one pipeline in Azure Data Factory runs only after another pipeline completes successfully?

21. What are parameters and variables in an Azure Data Factory pipeline, and how do they differ?

22. How can you perform parallel data copies in Azure Data Factory using partitioning?

23. How do you implement data partitioning in Azure Data Factory to improve pipeline performance?

24. How would you implement complex data transformations in Azure Data Factory involving filtering, aggregation, and joins from multiple source systems?

25. What are Global Parameters in Azure Data Factory, and when are they used?

26. How do you design and manage complex Azure Data Factory pipelines, and what challenges have you faced?

27. How can Azure Data Factory be used to orchestrate data movement and transformation across hybrid environments, including on-premises and cloud systems?

28. How would you implement incremental data loading from an on-premises SQL Server to Azure SQL Database using Azure Data Factory?

29. At which levels can parameterization be implemented in Azure Data Factory?

30. What are the commonly used activities in Azure Data Factory, and what are they used for?

31. How can you enforce a data retention policy using Azure Data Factory to meet data governance and compliance requirements?

32. How can you use Azure Data Factory to ingest and unify data from multiple file formats (CSV, JSON, Parquet) stored in Azure Data Lake Storage Gen2 into a single Azure SQL Database table?

33. What is the difference between annotations and user properties in Azure Data Factory, and how are they used?

34. How do you pass pipeline parameters from Azure Data Factory to an Azure Databricks notebook?

35. How do you monitor and troubleshoot Azure Data Factory pipelines to maintain reliable data processing?

36. How would you design an Azure Data Factory pipeline that dynamically chooses different source and destination systems based on runtime parameters such as file name, table name, or environment?

37. Is it mandatory to manually create an Integration Runtime (IR) in Azure Data Factory? Explain why or why not.

38. How can you securely connect Azure Data Factory to Salesforce?

39. How do you apply data masking in Azure Data Factory to safeguard sensitive data during processing?

40. How would you configure Azure Data Factory to ingest data from a REST API and load it into an Azure SQL Database on an hourly basis?

41. Can one Azure Data Factory pipeline invoke another pipeline? If yes, how is it implemented?

42. Where should sensitive credentials such as passwords or keys be securely stored in Azure Data Factory?

43. What is the role of Integration Runtimes in Azure Data Factory, and how do you decide which type to use?

44. When copying data from an on-premises database for multiple tables using a ForEach activity in Azure Data Factory, how can you ensure that only one database request is sent at a time?

45. How can you design a metadata-driven pipeline in Azure Data Factory to copy multiple tables from a source system to a target system?

46. How can Azure Data Factory be used to support near real-time processing for streaming data sources?

47. After validating an Azure Data Factory pipeline in the development environment, how would you promote it to UAT and production environments? What best practices do you follow?

48. How can you configure retry and error-handling mechanisms in Azure Data Factory pipelines?

49. How can you implement custom logging in Azure Data Factory to capture detailed pipeline execution information?

50. If an Azure Data Factory pipeline running in production suddenly stops working, how would you investigate and resolve the issue?

51. How can data validation and data quality checks be implemented in Azure Data Factory pipelines?

52. How can you implement data lineage tracking in Azure Data Factory to visualize and understand the flow of data through your pipelines?

53. If you need to pull data from an on-premises database to an Azure SQL Database daily in an incremental manner, how would you design the pipeline?

54. How can you optimize the performance of Azure Data Factory pipelines for faster and more efficient data processing?

55. How can you handle dynamic or evolving schema changes in source data when ingesting data using Azure Data Factory?

56. There is a business requirement where an external application drops files into Azure Blob Storage, and your pipeline needs to load the data into an Azure SQL Database. How would you design this solution in ADF?

57. How can you integrate Azure Data Factory with Azure Synapse Analytics for data movement and processing?

58. How can you enforce a data retention policy in Azure Data Factory to comply with organizational data governance standards?

59. You have a pipeline that copies data from source to destination. How would you set up an email notification to be sent automatically whenever the copy activity fails?

60. How can you handle schema drift in Azure Data Factory when the source schema changes dynamically?

61. How can you monitor and troubleshoot Azure Data Factory pipelines to ensure smooth and reliable data processing?

62. You are developing a daily pipeline that copies data from a source to Azure Data Lake Storage (ADLS). You need to organize the output files into a folder hierarchy by year/month/day. How would you implement this in ADF?

63. How can you integrate Azure Data Factory with Azure Databricks to perform complex data transformations?

64. How can you implement data masking in Azure Data Factory to secure sensitive information during data processing?

65. You have multiple files in an ADLS folder, where each file name corresponds to a table in a database (e.g., customer.txt, product.txt). How would you design a pipeline in ADF to automatically copy data from these files to their respective tables?

66. How can you implement data masking in Azure Data Factory to protect sensitive information during data processing?

67. How can you implement a data archiving solution in Azure Data Factory to efficiently manage historical data?

68. You are copying data from a file to a table using Azure Data Factory. Some rows do not match the table schema, causing the Copy activity to fail. How would you handle this scenario?

69. What are the steps to migrate existing SSIS packages to Azure Data Factory and execute them in the cloud?

70. How can you implement a dynamic pipeline in Azure Data Factory that processes multiple files with different or changing schemas?

71. You are copying data from a file to a table using Azure Data Factory, but the pipeline is performing very slowly. How can you improve its performance?

72. What is the role of Azure Data Factory in hybrid data integration scenarios involving both on-premises and cloud systems?

73. How can you improve the performance of data movement activities in Azure Data Factory when working with large datasets?

74. If a pipeline fails at the second activity, how can you rerun it from the failed activity in Azure Data Factory to avoid data inconsistency? Is it possible? Explain.

76. How would you design an Azure Data Factory pipeline to perform incremental data loads from an on-premises SQL Server to Azure Data Lake Storage, ensuring both data consistency and minimal latency?

77. How would you design an Azure Data Factory pipeline to perform incremental data loads from a source system that does not support Change Data Capture (CDC), while ensuring data consistency and minimal impact on the source system?

78. How would you design an Azure Data Factory solution to orchestrate a complex ETL workflow that includes conditional logic, robust error handling, and notifications, while ensuring maintainability?

79. How would you design an Azure Data Factory pipeline to read streaming data from Azure Event Hub, process it in near real-time, and store the results in an Azure SQL Database?

80. How would you design an Azure Data Factory pipeline that executes different data processing logic based on a parameter value passed at runtime?

AZURE DATABRICKS:

1. How can you read different types of files in Azure Databricks using Spark?

2. Can you explain the difference between a job cluster and an interactive cluster in Databricks?

DHANASEKARAN D
www.linkedin.com/in/dhanasekarand63

3. What is Azure Databricks, and how does it work with other services in the Azure ecosystem?

4. Can you explain what Change Data Capture (CDC) is and how it is used in data pipelines?

5. What are magic commands in Databricks, and when would you use them?

6. Can you explain the Databricks Lakehouse architecture and the purpose of the Bronze, Silver, and Gold layers?

7. Can you describe the key components of Azure Databricks architecture?

8. How does Azure Databricks ensure data security and maintain compliance?

9. Why is it not recommended to use inferred schema for large datasets in Databricks?

10. How do you implement data partitioning in Databricks, and when is it necessary to repartition data?

11. How can you execute one Databricks notebook from another notebook?

12. How can you copy files from Azure Blob Storage to Databricks?

13. What cluster modes are available in Azure Databricks, and when should you use each mode?

14. What techniques can you use to optimize the performance of a Spark job in Azure Databricks?

15. What is a prolonged secrets scope in Databricks, and how is it used?

16. How would you debug complex Spark applications in Azure Databricks?

17. How do you create a cluster in Azure Databricks?

18. How do you create and deploy notebooks in Azure Databricks?

19. What is Delta Lake, and how does it improve data reliability in Azure Databricks?

19. How do you manage version control and collaboration in Azure Databricks?

20. What parameters are required when mounting storage in Azure Databricks, and how should you handle credentials?

21. How can you implement a CI/CD pipeline for Azure Databricks notebooks?

22. How can you view the list of files in Databricks, both via commands and the UI?

23. What is Unity Catalog in Databricks, and what are its key features?

24. What are Databricks Jobs, and how are they used for workflow automation?

25. How does Azure Databricks work with Azure Machine Learning for model development and deployment?

26. Does Databricks support stored procedures, and how can you implement procedural logic?

27. How can you connect Azure Data Lake Storage (ADLS) to Databricks?

28. How can you implement data versioning in Azure Databricks to track historical changes?

29. What is the role of Azure Databricks in a modern data architecture?

30. How can you configure and manage job retries in Azure Databricks?

31. What is the difference between a Data Lake and Delta Lake in Azure Databricks?

32. What are Delta logs in Delta Lake, and how can you track data versioning in Delta tables?

33. How can you monitor and troubleshoot performance issues in Azure Databricks?

34. What factors should you consider when scaling Azure Databricks clusters for large datasets?

35. How can you securely manage secrets in Azure Databricks when accessing external data sources

36. What is a Hive Metastore, and what role does it play in Azure Databricks?

37. How can you implement SCD Type 1 and Type 2 using PySpark in Azure Databricks?

38. How can you manage dependencies and libraries in Azure Databricks notebooks?

39. How can Azure Databricks be integrated with Azure DevOps for CI/CD pipelines?

40. How can you mask or protect sensitive data (e.g., PII) before sharing it across teams in Azure Databricks?

41. If you have developed a Databricks notebook that needs to run every hour, how would you automate its execution?

42. How does auto-scaling work in Azure Databricks clusters, and why is it useful?

43. How can you handle schema evolution in Delta Lake when working with Azure Databricks?

44. What are the best practices for securing data and resources in Azure Databricks?

45. As a Data Engineer at a company, if you need to reuse code developed by another team in an Azure Databricks notebook, how would you do it following best practices?

46. What are some key optimization techniques you can apply in Azure Databricks to improve performance

47. How can you implement real-time or streaming data processing in Azure Databricks?

48. What are Databricks Delta Live Tables (DLT), and how do they simplify ETL processes in Azure Databricks?

49. How do you decide which cluster type and configuration to use in Azure Databricks for different workloads?

50. How can you monitor and track the performance of jobs in Azure Databricks?

51. How do you handle data skew in Azure Databricks to improve Spark job performance?

52. What are the key considerations for implementing role-based access control (RBAC) in Azure Databricks?

53. If multiple team members are working on the same Databricks notebook, how can you ensure that no one overwrites each other's work?

54. How can you grant permissions to specific notebooks or clusters for different users in Azure Databricks?

55. How do you implement incremental data processing in Azure Databricks for efficient ETL pipelines?

56. How would you implement a Slowly Changing Dimension (SCD) Type 2 in Azure Databricks using Delta Lake?

57. Can you call a Databricks notebook from Azure Data Factory (ADF)? If yes, how would you do it?

58. How do you implement data validation and quality checks in Azure Databricks?

59. If you have a Databricks notebook and are required to write the resultant DataFrame to Snowflake, how would you handle this?

60. How can you manage and automate ETL workflows using Azure Databricks Workflows?

61. What are the best practices for securing data in Azure Databricks?

62. If you create a Spark SQL table in Databricks and load data into it, but the cluster is terminated by mistake, will the table still be accessible? Explain why.

63. How can you integrate Azure Databricks with Azure DevOps for implementing CI/CD pipelines?

64. How can you use Databricks Delta Live Tables (DLT) for continuous data processing?

65. How can you stop or halt a Databricks process that is already running?

AZURE SYNAPSE ANALYTICS:

1. What advantages does Azure Synapse Analytics offer for handling large-scale data workloads?

2. How is Azure Synapse Analytics different from traditional data warehouse solutions?

3. In what ways does Azure Synapse Analytics work with other Azure services to support end-to-end data analytics?

4. What are the main components of Azure Synapse Analytics and their roles?

5. When would you opt for a dedicated SQL pool instead of a serverless SQL pool in Azure Synapse Analytics?

6. How can you monitor and optimize performance in Azure Synapse Analytics?

7. How can you implement data partitioning in Azure Synapse Analytics to enhance query performance?

8. How do you implement row-level security in Azure Synapse Analytics to control user access to data?

9. How can you perform real-time analytics on streaming data using Azure Synapse Analytics?

10. What are control nodes and compute nodes in Azure Synapse Analytics, and how do they collaborate during query execution?

11. How is data ingested into Azure Synapse Analytics, and what are the typical sources and storage options?

12. What are materialized views in Azure Synapse Analytics, and how do they improve query performance?

13. How is data secured in Azure Synapse Analytics through encryption at rest and in transit?

14. How can Azure Synapse Analytics be integrated with Power BI for reporting and visualization?

15. Can you describe a scenario where PolyBase would be used in Azure Synapse Analytics?

16. What are the best practices for data partitioning and distribution in Azure Synapse Analytics?

17. How do you implement a Slowly Changing Dimension (SCD) Type 2 in Azure Synapse Analytics to track historical changes?

18. Which tools would you use to schedule and orchestrate data pipelines in Azure Synapse Analytics?

19. How can you create a dedicated SQL pool in Azure Synapse Analytics, and what scenarios is it best suited for?

20. How can you reduce storage costs in Azure Synapse Analytics when managing large datasets?

21. How can you implement security and ensure compliance in Azure Synapse Analytics?

22. How can you implement data masking in Azure Synapse Analytics to protect sensitive information?

23. What steps would you follow to migrate an on-premises data warehouse to Azure Synapse Analytics?

24. What are the different types of data distribution in Azure Synapse SQL Pools, and when would you use each?

25. Can you describe a scenario where you would use serverless SQL pools in Azure Synapse Analytics?

26. How can you implement data versioning in Azure Synapse Analytics to track historical changes?

27. How can you implement parameterization in Azure Synapse pipelines to enhance reusability and flexibility?

28. How would you design a solution in Azure Synapse Analytics to process real time streaming data with complex aggregations?

29. How would you design and implement a data lakehouse architecture using Azure Synapse Analytics?

30. What types of activities can you use in Azure Synapse pipelines, and in which scenarios would you use them?

AZURE SQL DATABASE:

1. What is the role of Azure SQL Database in cloud-based applications, and what benefits does it offer?

2. What are the key differences between Azure SQL Database and Azure SQL Managed Instance?

3. How do you set up an Azure Data Factory (ADF) pipeline for ETL processes?

4. How can you implement row-level security (RLS) in Azure SQL Database to control user-specific data access?

5. What are the best practices for securing an Azure SQL Database?

6. How can you monitor and improve query performance in Azure SQL Database?

7. How is data partitioning implemented in Azure SQL Data Warehouse (Synapse Analytics), and what are the best practices?

8. What steps would you follow to migrate an on-premises SQL Server database to Azure SQL Database?

9. What are the key differences between Azure Cosmos DB and a traditional SQL database?

10. How can you implement geo-replication in Azure SQL Database to ensure disaster recovery?

11. How can Azure SQL Database be integrated with other Azure services, and what are the typical use cases?

12. How can you secure sensitive data in Azure SQL Database without modifying the application code?

13. How can Azure Data Lake Storage (ADLS) be integrated with SQL-based systems for analytics and ETL?

14. How can you implement automatic scaling for an Azure SQL Database based on workload requirements?

15. How can Azure Logic Apps be used to automate tasks in Azure SQL Database?

16. How can you implement an effective index maintenance strategy in Azure SQL Database?

17. What are the steps to migrate an on-premises SQL Server database to Azure SQL?

18. How can you monitor and manage resource utilization in Azure SQL Database?

19. What is U-SQL in Azure Data Lake Analytics, and what are its typical use cases?

20. How can you handle sudden spikes in database traffic in Azure SQL Database?

21. How can you ensure high availability and disaster recovery in Azure SQL Database?

22. How can you monitor and troubleshoot deadlocks in Azure SQL Database?

23. What are the common challenges in managing large-scale SQL databases in Azure?

24. How can you audit and track changes to specific tables in Azure SQL Database?

25. How do you set up and manage an Azure Synapse Analytics workspace?

26. How would you optimize a query with multiple joins and aggregations in Azure SQL Database?

27. How can you monitor and troubleshoot Azure SQL Database effectively?

28. How can you implement data archiving in Azure SQL Database to efficiently manage large datasets?

29. How can Azure Active Directory (AAD) be used for authentication in Azure SQL Database, and what are the benefits?

30. How can you implement data encryption in Azure SQL Database to secure data at rest, in transit, and at the column level?

31. How can Azure SQL Data Sync be used for replicating data across databases, and what are its key components?

32. What are the key considerations for designing a scalable data architecture in Azure?

AZURE DEVOPS AND CI/CD:

1. How can you implement CI/CD pipelines using Azure DevOps for deploying applications and services?

2. How can CI/CD practices improve the reliability and consistency of data pipelines in data engineering?

3. What are the main components and key features of Azure DevOps?

4. How can you manage dependencies between multiple data pipelines in a CI/CD workflow?

5. What is the purpose of pipelines in Azure DevOps, and how are they used in software delivery?

6. How do you monitor and track the performance of data pipelines after deployment in a CI/CD workflow?

7. How do you handle dependencies and versioning in Azure DevOps pipelines?

AZURE FUNCTION:

1. What types of triggers can Azure Functions use, and how do they work?

2. What are Azure Functions, and how do they differ from traditional web services or APIs?

3. What are input and output bindings in Azure Functions, and how do they differ?

4. How do you deploy and manage Azure Functions in production environments?

5. How does Azure Functions handle parallel execution of multiple requests or events?

6. What are some common use cases for Azure Functions in data engineering workflows?

7. How do you implement CI/CD pipelines for deploying Azure Data Factory and Databricks solutions?

AZURE KEY VAULT:

1. What is Azure Key Vault and why is it used?

2. How can you securely manage secrets, keys, and certificates using Azure Key Vault?

3. What is the difference between secrets, keys, and certificates in Azure Key Vault?

4. How can Azure Key Vault be integrated with other Azure services for secure secret management?

5. How can Azure Monitor be used to track and analyze the performance of Azure services?

DATA MODELLING:

1. How would you design a simple star schema data warehouse for a grocery store?

2. What are the different types of data warehouse schemas, and how do you decide which one to use for a data model?

3. What are fact tables and dimension tables in data modeling, and how are they used?

4. What is the difference between a database, a data warehouse, and a data lake, and when would you use each?

5. What is database normalization, and why is it important in data management?

6. What is the difference between ER (Entity-Relationship) modeling and dimensional modeling in data systems?

7. What is denormalization in database design, and when is it appropriate to use?

8. How do you handle Slowly Changing Dimensions (SCD) in data modeling, and what are the different types?

9. What are surrogate keys in dimensional modeling, and why are they used?

10. How do you optimize data models for performance and scalability in data warehouses?

11. What is a galaxy schema in data warehousing, and when is it used?

12. What is the role of metadata in data modeling and data architecture?

13. What is a data mart, and how does it differ from a data warehouse?

14. What is a junk dimension, and when would you use it?

15. What is a factless fact table, and why is it used?

DHANASEKARAN D
www.linkedin.com/in/dhanasekarand63

SQL:

1. What is the difference between DELETE and TRUNCATE commands in SQL

2. How does a Primary Key differ from a Unique Key in SQL?

3. What are Views and Common Table Expressions (CTEs), and how do they differ?

4. How do DELETE TABLE and TRUNCATE TABLE differ in SQL?

5. What are the key differences between Stored Procedures and Functions in SQL?

6. How does the HAVING clause differ from the WHERE clause in SQL?

7. What is the difference between aggregation functions and analytic (window) functions in SQL

8. What are the differences between Star Schema, Snowflake Schema, and Third Normal Form (3NF)?

9. Can you explain the ACID properties of transactions in SQL with an example?

10. How do you create and use clustered and non-clustered indexes in SQL?

11. What is the difference between UNION and UNION ALL in SQL?

12. What are common concurrency issues in SQL databases, and how can they be handled?

13. What are constraints in SQL?

14. Difference between DELETE, DROP, and TRUNCATE?

15. Difference between EXISTS and IN?

16. What are isolation levels?

17. What are different types of joins?

SQL CODING QUESTIONS:

1. How can you find the second-highest salary in an employee table using SQL?

2. How can you retrieve the top 3 highest-paid employees in each department using SQL

3. How can you retrieve the Nth highest salary from an employees table using SQL?

DHANASEKARAN D
www.linkedin.com/in/dhanasekarand63

4. How do you identify and count duplicate records in a table, for example duplicate employee names?

5. How can you retrieve records that are present in one table but missing in another table?

6. How can you retrieve employees who earn more than the company's average salary?

7. find employees whose salary is higher than their manager's salary?

8. Remove duplicate records from a table while keeping only one unique row?

9. Identify the top three highest-paid employees in each department?

10. How can you retrieve records that exist in both tables

11. Transform row-based data into column-based data using a PIVOT operation?

12. How can you fetch the most recent 10 records from a table?

13. Retrieve rows that exist in one table but do not exist in another table

14. How can you retrieve the five highest-paid employees from a table?

15. Calculate a running (cumulative) total of salaries within each department?

16. Retrieve employees who were hired in a specific year (for example, 2025)

17. Find customers who made purchases in the previous year but have not made any purchases in the current year?

18. Retrieve employees who have joined within the last six months?

19. Identify the department that has the maximum number of employees

20. Identify customers who have purchased every product from a specific list of products?

21. Find the highest salary in each department?

22. check number of employees working in each department?

23. consecutive absence days for employees using attendance data?

24. Join two related tables and retrieve only selected columns from each

25. Retrieve employees who earn the highest salary within their respective departments

26. Find orders that have been dispatched but have not yet been invoiced?

27. Identify missing numbers in a continuous numeric sequence stored in a table

28. Retrieve the youngest employee in an organization?

29. Performance-based rankings to employees within each department
30. calculate the average salary for each department
31. Retrieve the earliest and most recent records from a table
32. Identify the earliest purchase made by each customer
33. Calculate the median salary of employees using SQL
34. identify the department that has the lowest average salary
35. Calculate the average sales for each month of every year
36. find all employees who do not have a manager assigned?
37. Retrieve the second-highest value from a column in a table
38. Calculate a running total of sales over time using SQL
39. Create a new table from the results of a query in SQL
40. Retrieve all employees who do not manage any subordinates
41. Find products that appear in multiple orders with the same price
42. check if a table contains any rows or is empty using SQL
43. Retrieve all orders placed within the last 7 days?
44. Total number of employees for each job title
45. Retrieve the highest-value order for each customer
46. Find employees who are assigned to both departments 101 and 102?
47. Find the best-selling product for each month?
48. Retrieve all employees who share the same salary with at least one other employee
49. Calculate customer cohort retention rate using SQL
50. Update employee salaries differently based on their department in a single query

PYSPARK:

1. What are the advantages of using Apache Spark over Hadoop MapReduce?
2. Can you explain what a DAG is in Spark and how Spark uses it to execute tasks?

DHANASEKARAN D
www.linkedin.com/in/dhanasekarand63

3. What are the key components of the Apache Spark ecosystem and their purposes?

4. Can you explain the difference between Apache Spark, Databricks, and PySpark?

5. Can you explain the Apache Spark architecture, specifically the roles of the Driver and Worker nodes?

6. What is the difference between SparkSession and SparkContext in Apache Spark?

7. Can you explain the difference between transformations and actions in Apache Spark?

8. What is the difference between narrow and wide transformations in Apache Spark, and why does it matter?

9. What is meant by lazy transformations in Apache Spark, and why are they important?

10. What strategies can be used to improve the performance of a Spark job?

11. What is the difference between a DAG and lineage in Apache Spark?

12. Can you explain the difference between RDDs and DataFrames in Apache Spark?

13. What is the difference between groupByKey() and reduceByKey() in Spark, and when should each be used?

14. What is PySpark, and how does it enable Python applications to run on Spark?

15. Can you explain the Spark RDD transformations distinct(), union(), intersection(), and subtract()?

16. What are partitions in Apache Spark, and why are they important?

17. What is the difference between repartition() and coalesce() in Apache Spark, and when should each be used?

18. What are the main characteristics of PySpark?

19. What is the difference between mapPartitions() and mapPartitionsWithIndex() in Spark?

20. What is data shuffling in Spark, and why is it considered an expensive operation?

21. What does inferSchema do when reading a file in Spark, and when should it be used?

22. What is a broadcast join in PySpark, and when should it be used?

23. How do you register and use a User Defined Function (UDF) in PySpark?

24. What are the key features of Apache Spark, and what are the advantages and disadvantages of using PySpark?

25. What are some commonly used transformations in PySpark, and how do they work?

26. Why are broadcast variables needed in Apache Spark, and how do they work?

27. How does caching work in PySpark, and when should it be used?

28. What is the difference between repartition() and coalesce() in Spark, and when should each be used?

29. What is the difference between map() and flatMap() transformations in PySpark?

30. What are the different ways to define a schema in PySpark, and why is it recommended to define one?

31. Can you show a PySpark example that reads CSV files, joins two DataFrames, performs an aggregation, and filters rows based on a condition?

32. What is the difference between cache() and persist() in PySpark?

33. How do you handle skewed data in Spark, and what techniques can you use to balance the load?

34. What is Spark SQL, and how is it used for structured data processing?

35. How can you create a rank column in PySpark using Window functions, and what are the alternatives?

36. What is the difference between broadcast variables and accumulator variables in Spark?

37. What is the Catalyst Optimizer in PySpark, and how does it work?

38. How do you create a temporary view in PySpark, and what are its characteristics?

39. What is Z-ordering in Spark Delta Lake, and how does it improve query performance?

40. What is Delta Lake, and what advantages does it offer over plain Parquet files?

41. What is the difference between collect(), collectAsList(), and select() in PySpark?

42. How can you handle null values in PySpark DataFrames, and what are the common approaches?

43. How can you create a global temporary view in PySpark, and what are its characteristics?

44. What is Adaptive Query Execution (AQE) in Databricks Spark, and how does it improve query performance?

45. Why are DataFrames faster than RDDs in Spark, and how do they compare with Datasets?

46. What is the purpose of checkpointing in Spark, and when should it be used?

47. How can you handle incremental loads in PySpark when a table does not have a last_modified or updated_time column?

48. How are jobs, stages, and tasks structured during a Spark job execution?

49. How do you perform pivot and unpivot operations on a Spark DataFrame

50. What are accumulators in PySpark, and in what scenarios would you use them?

51. What is the difference between Spark SQL and PySpark DataFrame APIs, and when would you use each?

52. How do you filter and aggregate data in PySpark using the DataFrame API? Provide an example.

53. How do you handle schema evolution in Azure Data Lake using Delta Lake?

54. Is PySpark faster than Pandas, and why would you choose PySpark over Pandas?

55. What is the purpose of selectExpr in PySpark DataFrames, and when would you use it?

56. What is data skew in Spark, and why is it a problem?

57. How would you troubleshoot a PySpark job that fails due to memory issues?

58. What are some common optimization techniques in Apache Spark to improve job performance?

59. How would you design and implement an ETL pipeline using PySpark?

60. What is salting in Spark, and how is it used to handle data skew?

61. What are the different Spark deployment modes, and how do Cluster and Client modes differ?

62. How can you remove duplicate records from a PySpark DataFrame while keeping only the first occurrence?

63. How would you calculate the average salary and the number of employees in each department using PySpark?

64. Given a PySpark DataFrame of sales data, how would you identify the top 5 products based on total sales volume?

65. How would you filter a PySpark DataFrame to include only rows where salary is greater than 5000 and select just the name column?

66. How can you compute a cumulative sum of a column in PySpark for each department?

67. How do you perform inner, left, and right joins on two PySpark DataFrames?

68. How would you find employees in PySpark who joined within the last six months?

69. How would you perform a right join in PySpark to get all customers, including those who haven't placed any orders?

70. How do you join two PySpark DataFrames using multiple columns as the join condition?

71. How can you rename columns in a PySpark DataFrame?

72. How would you calculate the salary difference between consecutive employees within each department in PySpark?

73. How can you sort a PySpark DataFrame based on one or more columns?

74. If a DataFrame column contains comma-separated values, how can you split it into multiple individual columns in PySpark?

75. How can you rank employees within each department based on their salary using PySpark?

76. How can you calculate a running total (cumulative sum) of a column for each category in a PySpark DataFrame?

77. How can you reduce the number of partitions in a PySpark DataFrame using coalesce()?

78. How can you retrieve the nth highest salary from a PySpark DataFrame?

79. What are the different methods to read data into a PySpark DataFrame?

80. How can you pivot a PySpark DataFrame based on a category column and aggregate values?

DHANASEKARAN D
www.linkedin.com/in/dhanasekarand63

PYTHON:

1. What are the main differences between Python lists, tuples, and sets in terms of mutability, ordering, duplicates, and typical use cases?

2. What is the difference between local and global variables in Python, and how does their scope affect program behaviour?

3. How do you detect and handle missing or null values in a dataset using Python (pandas)?

4. What is the difference between a class method and a static method in Python, and when would you use each?

5. What is data serialization and deserialization in Python, and how can you perform it using JSON and Pickle?

6. What is the difference between a shallow copy and a deep copy in Python, and when would you use each?

7. How can you read from and write data to a database in Python? Provide examples using SQLAlchemy and pandas.

8. What techniques can you use to optimize Python code for better performance?

9. What are Python list comprehensions, and how can you use them to create lists efficiently? Provide an example with a condition.

10. How do you handle exceptions in Python, and how can you raise custom exceptions? Provide examples.

11. What are context managers in Python, and how are they used for resource management? Provide examples.

12. How can Python be integrated with big data tools like Apache Spark, and what are common use cases?

13. What is the role of the pandas library in Python for data manipulation, and what are its key features? Provide a short example.

14. What are Python's key features?

15. Explain Python decorators

16. Explain Python's exception handling

17. How to integrate Python with Apache Spark

18. Explain Python's context managers

DHANASEKARAN D
www.linkedin.com/in/dhanasekarand63

PYTHON CODING QUEATIONS:

1. Write a Python generator to produce an infinite Fibonacci sequence.

2. Write a Python generator to produce prime numbers indefinitely.

3. Write Python code to sort a list without using the built-in sort() method.

4. Write Python code to check whether a given string is a palindrome.

5. Write Python code to sort a list using the built-in sort() method.

6. Write a Python function to replace all vowels in a string with spaces.

7. Write Python code to verify if a given string is a palindrome.

8. Count the number of occurrences of each word in a given string.

9. Write Python code to sort a dictionary by its keys.

10. Find the first number in a list that appears consecutively three times.

11. Find all pairs of numbers in a list that sum up to a given target number.

12. Write Python code to split a full name string into first name and last name.

13. Write Python code to find the character that appears most frequently in a string. from collections import Counter

14. Write Python code to find duplicate elements in a list and count how many times each occurs.

15. How can dictionaries be used in Python to efficiently store and access key value pairs? Provide examples and tips.

16. Write a Python program to calculate the factorial of a given number.

17. Write Python code to load a CSV file into a pandas DataFrame and display the f irst 5 rows.

18. Write Python code to check whether a given number is an Armstrong number.

19. Write Python code to reshape a 1D NumPy array into a 2D array.

20. Write Python code to filter rows in a pandas DataFrame where a specific column value exceeds a given threshold (e.g., age > 30).

DHANASEKARAN D
www.linkedin.com/in/dhanasekarand63

21. Write Python code to get the ASCII value of a character and convert an ASCII value back to a character.

22. Write Python code to find the indices of all non-zero elements in a NumPy array

23. Write Python code to add a new column to a pandas DataFrame that is calculated from existing columns (e.g., total = price × quantity).

24. Write Python code to calculate the sum of squares of the first n natural numbers.

25. Write Python code to solve a system of linear equations using NumPy.

26. Write Python code to drop all rows in a pandas DataFrame that have missing (NaN) values in any column.

27. Write Python code to find all common letters between two strings.

28. Write Python code to group a pandas DataFrame by a categorical column and calculate the average of a numerical column.

29. Write Python code to create a dictionary by combining two lists: one for keys and one for values.

30. Write Python code to perform element-wise division of two NumPy arrays

31. Write Python code to transform a pandas DataFrame from wide format to long format using melt().

32. Write Python code to reverse a given string.

33. Write Python code to remove duplicate rows from a pandas DataFrame while keeping the first occurrence.

34. Write Python code to count the number of words in a paragraph.

35. Write Python code to replace values in a pandas DataFrame column using a dictionary mapping with the map() function.

36. Write Python code to find the minimum and maximum values in a NumPy array.

37. Write Python code to merge two strings by alternating their characters.

38. Write Python code to filter rows in a pandas DataFrame where a timestamp column falls between two specific dates.

39. Write Python code to find the greatest common divisor (GCD) of two strings, i.e., the largest string that can be concatenated multiple times to form both strings.

40. Write Python code to reverse only the vowels in a given string.

DHANASEKARAN D
www.linkedin.com/in/dhanasekarand63

41. Write Python code to merge multiple pandas DataFrames and calculate aggregations (e.g., sum or mean) on a column.

42. Write Python code to move all zeros in a list to the end while maintaining the order of non-zero elements.

43. Write Python code to check if one string is a subsequence of another string.

44. Write Python code to flatten a nested JSON object into a flat dictionary or DataFrame.

45. Write Python code to find the unique elements in a list and count how many times each element occurs.